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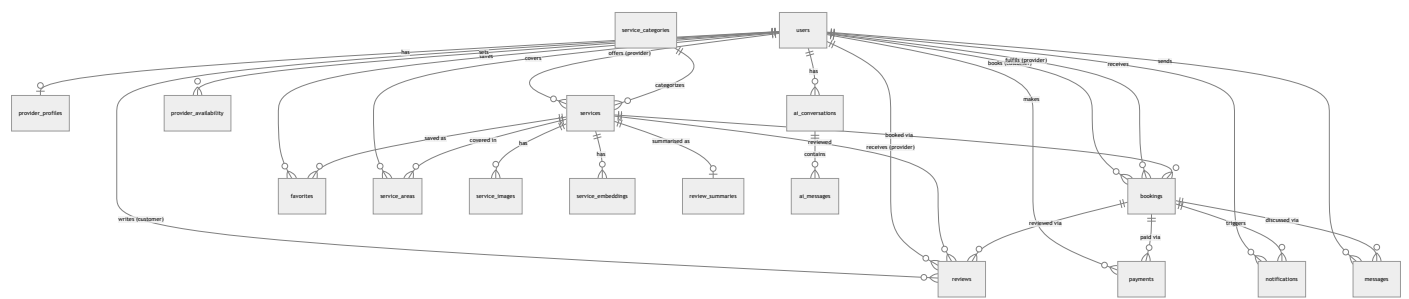
Database Design

Entity-relationship diagram, table dictionary, and integrity rules for the live schema.

Program	Diploma in Advanced AI & Software Engineering
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1. Entity-Relationship Diagram

The live schema (PostgreSQL, hosted on Supabase) has 17 domain tables. Relationships are enforced with real foreign keys, not just application logic.



2. Table Dictionary

users

Every account — customer, provider, or admin — in one table, distinguished by `role`.

COLUMN	TYPE	NOTES
<code>id</code>	integer, PK	Auto-increment
<code>email</code>	varchar, unique	Login identifier
<code>password_hash</code>	varchar	bcrypt hash; never returned by the API
<code>full_name</code>	varchar	
<code>phone</code>	varchar, nullable	
<code>role</code>	enum	customer / provider / admin
<code>status</code>	enum	active / pending / suspended / deactivated
<code>email_verified</code>	boolean	
<code>created_at</code> / <code>updated_at</code>	timestampz	

provider_profiles

One-to-one extension of `users` for provider-only fields.

COLUMN	TYPE	NOTES
user_id	integer, PK, FK → users.id	
bio	text, nullable	
years_experience	integer	
verification_status	enum	unverified / pending / verified / rejected
avg_rating	numeric	Recalculated whenever a review is created
total_reviews	integer	Recalculated whenever a review is created

service_categories

Marketplace categories (Plumbing, Electrical, Cleaning, Tutoring, Repairs, Tech Support, and any admin adds).

COLUMN	TYPE	NOTES
id	integer, PK	
name	varchar, unique	
description	text, nullable	
icon	varchar	Emoji shown in category pills

services

A provider's individual listing.

COLUMN	TYPE	NOTES
id	integer, PK	
provider_id	integer, FK → users.id	
category_id	integer, FK → service_categories.id	
title / description	varchar / text	
price	numeric	LKR
city	varchar	
duration	varchar	e.g. "1-2 hours"
status	enum	active / paused / draft

bookings

The core transaction: a customer hiring a provider for a service.

COLUMN	TYPE	NOTES
id	integer, PK	
customer_id / provider_id	integer, FK → users.id	Both reference the same table
service_id	integer, FK → services.id	
booking_date	timestamptz	Must be in the future on creation
status	enum	pending / accepted / rejected / in_progress / completed / cancelled
notes	text, nullable	

reviews

One review per completed booking.

COLUMN	TYPE	NOTES
id	integer, PK	
booking_id	integer, FK → bookings.id, unique	Enforces one review per booking
customer_id / provider_id / service_id	integer, FK	Derived from the booking server-side
rating	integer	1–5
comment	text, nullable	

payments

Simulated payment record for a completed booking.

COLUMN	TYPE	NOTES
id	integer, PK	
booking_id	integer, FK → bookings.id	
customer_id	integer, FK → users.id	
amount	double precision	
status	varchar	pending / completed / failed / refunded
payment_method	varchar	card / bank / cash
transaction_id	varchar	Generated on success

messages

Direct messages, scoped to one booking's two participants.

COLUMN	TYPE	NOTES
id	integer, PK	
booking_id	integer, FK → bookings.id	
sender_id	integer, FK → users.id	
body	text	
is_read	boolean	

notifications

In-app notifications for booking, review, and message events.

COLUMN	TYPE	NOTES
id	integer, PK	
user_id	integer, FK → users.id	Recipient
title / message	varchar / text	
is_read	boolean	
booking_id	integer, FK → bookings.id, nullable	

provider_availability

A provider's recurring weekly open hours.

COLUMN	TYPE	NOTES
id	integer, PK	
provider_id	integer, FK → users.id	
day_of_week	integer	0 = Sunday .. 6 = Saturday
start_time / end_time	time	

favorites

A customer's saved services.

COLUMN	TYPE	NOTES
id	integer, PK	
user_id	integer, FK → users.id	
service_id	integer, FK → services.id	

service_areas / service_images / service_embeddings

Supporting tables: coverage districts per provider, listing photos, and vector embeddings for AI search.

COLUMN	TYPE	NOTES
service_areas	district, city, radius_km	FK → users.id (provider), services.id (optional)
service_images	image_url	FK → services.id
service_embeddings	embedding (text)	FK → services.id

ai_conversations / ai_messages

Chat history for the AI assistant, one conversation per thread.

COLUMN	TYPE	NOTES
ai_conversations	user_id, title, timestamps	FK → users.id
ai_messages	conversation_id, role, message	FK → ai_conversations.id

review_summaries

Cached AI-generated summary of a service's reviews.

COLUMN	TYPE	NOTES
id	integer, PK	
service_id	integer, FK → services.id	
summary	text	Regenerated on request

3. Normalization

The schema is in Third Normal Form (3NF): every non-key column depends only on its table's primary key, repeating groups are split into their own tables (e.g. `service_images` , `service_areas` instead of array columns), and derived values that must stay in sync (a provider's `avg_rating` / `total_reviews`) are recalculated on write rather than trusted as user input.

4. Key Constraints & Integrity Rules

- A `reviews` row is unique per `booking_id` — the database itself prevents a second review on the same booking.
- All foreign keys use `ON DELETE CASCADE` from `users` , so removing an account cleans up its bookings, services, and messages consistently.
- Enum columns (`role` , `status` , `booking_status` , `verification_status`) are backed by real Postgres enum types, not free-text strings.